CI1 Cybersecurity Engineering Roadmap

Hands-On SOC Lab Portfolio & Strategic Execution Plan

Target: Cycle Ingénieur 1 (2026–2027) ⟶ M2 / CI3 Cybersecurity Alternance Preparation

1. Portfolio Ranking & Implementation Matrix

Projects are sequenced hierarchically from foundational automation scripting to enterprise-grade telemetry and Active Directory defense. Each tier supplies tools, decoders, or workflows directly into subsequent tiers.

Rank	Project Title	Core Competencies	Effort	Hardware	Schedule
#1	Threat Intel Triage Tool	Python, Async APIs, JSON, CLI	Low	2 GB RAM	Oct 2026
#2	NIDS & Traffic Forensics	Suricata, Wireshark, Zeek, PCAP	Mod.	4–8 GB RAM	Nov 2026
#3	Automated SOC/SOAR Stack	Wazuh, Shuffle, TheHive, Sysmon	High	12–16 GB RAM	Jan–Feb 2027
#4	Active Directory Detection	Win Server, Splunk, Atomic Red	Adv.	16+ GB RAM	Mar–Apr 2027

2. Detailed Project Specifications

2.1 Rank 1: Automated Threat Intel Triage CLI/API Tool

Project 1 Architecture & Objective

Objective: Construct a modular Python utility and FastAPI service that accepts raw Indicators of Compromise (IOCs: IPs, domains, SHA-256 hashes), queries public threat feeds asynchronously, and calculates a normalized triage risk score.

- **Stack:** Python 3.11+, FastAPI, HTTPX/Requests, Typer/Click, Docker.
- **Integrated Feeds:** VirusTotal API v3, AbuseIPDB v2, AlienVault OTX, URLhaus.
- **Step-by-Step Implementation:**
  1. Develop regex-based input parsers distinguishing IPv4/IPv6, MD5, SHA-256, and FQDNs.
  2. Construct asynchronous API client classes with exponential backoff and rate-limit management.

3. Implement a scoring engine assigning weights based on reputation thresholds (e.g., AbuseIPDB confidence score $> 75\%$, VT positives ≥ 3).
4. Expose endpoints via FastAPI and provide a localized CLI displaying results formatted with the `Rich` terminal library.

- **French CV Description:**
  - ◇ *Développement d'un outil CLI/API (Python, FastAPI) d'enrichissement automatisé d'IOCs via les APIs VirusTotal et AbuseIPDB.*
  - ◇ *Implémentation d'un algorithme de scoring de réputation et génération de rapports d'investigation L1 au format JSON/PDF.*

2.2 Rank 2: Network Traffic & Intrusion Detection (NIDS) Lab

Project 2 Architecture & Objective

Objective: Deploy an inline/passive network intrusion detection system (NIDS), craft custom detection signatures, and perform forensic payload extraction from packet captures (PCAP).

- **Stack:** Suricata NIDS, Zeek, Wireshark, `tshark`, Brim/Zui, Linux Virtual Bridge.
- **Step-by-Step Implementation:**
  1. Install and configure Suricata on a Linux VM listening on a mirrored internal network interface.
  2. Write custom Suricata detection rules (`.rules`) detecting unauthorized SSH egress tunnels, base64 User-Agents, and raw SYN port scans.
  3. Ingest malicious PCAP samples (C2 beaconing from Cobalt Strike/Sliver, DNS exfiltration tunnels).
  4. Reconstruct TCP/TLS handshakes, extract suspicious binary payloads, and isolate extracted malicious certificates using Zeek logs (`conn.log`, `dns.log`, `ssl.log`).

- **French CV Description:**
  - ◇ *Déploiement et réglage fin d'une sonde NIDS Suricata : écriture de règles personnalisées et élimination des faux positifs.*
  - ◇ *Analyse médico-légale de flux réseau (PCAP) avec Wireshark et Zeek : détection de balises C2 et exfiltration DNS.*

2.3 Rank 3: End-to-End Automated SOC & SOAR Pipeline

Project 3 Architecture & Objective

Objective: Build a containerized enterprise Security Operations Center stack automating the full detection-to-containment workflow: log aggregation, signature alerting, SOAR enrichment, and ticket escalation.

- **Stack:** Wazuh SIEM/XDR, Shuffle SOAR, TheHive 5, Sysmon (SwiftOnSecurity config), Docker Compose.
- **Data Pipeline Architecture:**

$$Host(Sysmon) \longrightarrow WazuhAgent \longrightarrow WazuhManager Webhook Shuffle SOAR Enrichment TheHive 5$$

- **Step-by-Step Implementation:**
  1. Deploy Wazuh Indexer, Dashboard, Manager, Shuffle, and TheHive 5 via a unified `docker-compose.yml`.
  2. Deploy Wazuh Agents on a target Windows 11 workstation with Sysmon and an Ubuntu Server.
  3. Author custom Wazuh rules (`local_rules.xml`) and decoders to trigger alerts on Event ID 1 (suspicious process spawning from `svchost.exe` or `cmd.exe`).
  4. Configure Shuffle SOAR to intercept alerts, parse the suspect IP/hash, query the Rank 1 Threat Intel tool or VirusTotal API, and submit an enriched case directly to TheHive 5.
- **French CV Description:**
  - ◇ *Architecture d'un SOC conteneurisé (Wazuh SIEM/XDR, Shuffle SOAR, TheHive 5) supervisant un parc Windows (Sysmon) et Linux.*
  - ◇ *Automatisation du cycle de triage SOAR : déclenchement par webhook, enrichissement de télémétrie et création automatisée d'incidents.*
  - ◇ *Conception de décodeurs et de règles de détection XML personnalisées sous Wazuh.*

2.4 Rank 4: Active Directory Attack & Detection Lab (MITRE ATT&CK)

Project 4 Architecture & Objective

Objective: Configure an Active Directory domain environment, execute controlled red-team attacks using Atomic Red Team, and engineer centralized detection rules in Splunk using the MITRE ATT&CK framework.

- **Stack:** Windows Server 2022 (Domain Controller), Windows 10 Enterprise, Splunk Enterprise, Atomic Red Team, Sigma.
- **Step-by-Step Implementation:**
  1. Deploy an AD domain (`CORP.LOCAL`) with organizational units, administrative accounts, and Service Principal Names (SPNs).
  2. Configure advanced audit policies (Kerberos Service Ticket Operations, Process Creation with command lines, PowerShell Script Block Logging 4104).
  3. Install Splunk Universal Forwarder to aggregate security telemetry into Splunk Enterprise.
  4. Execute red-team simulations via Atomic Red Team:

  - *Kerberoasting* (T1558.003): Querying SPN tickets with RC4 encryption (Event ID 4769).
  - *Credential Dumping* (T1003): Accessing LSASS memory via ProcDump or comsvcs.dll.
5. Author production-ready detection queries in Splunk SPL and export rules in cross-platform **Sigma** format.

- **French CV Description:**
  - ◇ *Déploiement d'un environnement Active Directory audité avec supervision centralisée sous Splunk Enterprise.*
  - ◇ *Simulation d'attaques avec Atomic Red Team (Kerberoasting, extraction LSASS) et cartographie MITRE ATT&CK.*
  - ◇ *Développement de requêtes de détection SPL et formalisation de règles de sécurité au format Sigma.*

3. Execution Calendar: Cycle Ingénieur 1 (2026–2027)

Phase / Month	Primary Milestone	Key Technical & Academic Deliverables
Oct 2026	Project #1 (Python Tool)	Complete async API triage utility. Upload documented repository to GitHub. Lock down top grades in S5 math and networking.
Nov–Dec 2026	Project #2 (NIDS Lab)	Deploy Suricata. Analyze C2/DNS PCAP samples. Write formal investigation writeups with packet traces.
Jan 2027	S5 Final Exams	Validate S5 with top-tier class rank. Set up Docker host for Project #3.
Feb–Mar 2027	Project #3 (SOC/SOAR)	Deploy Wazuh, TheHive, and Shuffle. Test end-to-end incident handling pipeline.
Apr–May 2027	Project #4 (AD / Splunk)	Build Active Directory domain. Run Atomic Red Team simulations. Write Sigma and SPL detection rules.
Jun–Jul 2027	Profile Consolidation	Finalize S6 ranking. Structure 1-page French CV. Publish GitHub repositories with architecture diagrams.
Aug 2027	Technical Internship	Complete summer internship. Grind TryHackMe SOC Level 1 / Hack The Box paths. Prepare for CI2 application launch.